DAY 36 · THEORY HANDOUT

Give a Model a Small Python Tool

Define a tool name, description, input schema, and result.

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The idea and its useful terms

Define a tool name, description, input schema, and result.

Tool

A function the app lets a model request.

Tool schema

The name, purpose, and allowed inputs.

Decorator

The @tool line wraps a function as a tool.

A language model predicts text. A calculator function computes a number using Python. We can use each for the part it does well.

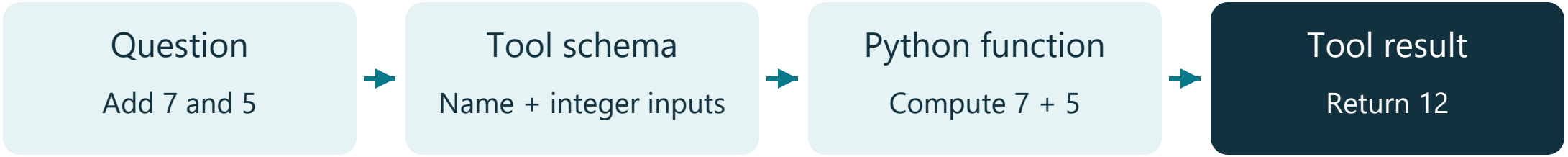
The description helps the model choose a tool. Type labels tell it what arguments to supply and let the app check them.

> A tool description should say when to use it. The model does not run Python just by naming a function.

Follow the work, one step at a time

Each step has a job. We can inspect its output before moving on.

Read the path from left to right.



Each arrow means: pass the result to the next step.

The function name becomes a tool name. Its docstring describes when to use it. Type annotations describe its arguments and result. @tool packages these details so the model can request the function.

Invoke the tool directly with known arguments first. That tests Python independently of model choice. When a model uses it later, the application still has to accept the requested name and validated arguments.

> If the answer is wrong, inspect the input and the intermediate results. The last sentence alone cannot show which step failed.

A small example we can check

The values here are classroom examples. The notebook shows actual results.

Name	Inputs	Result
add_numbers A clear action name.	a: integer, b: integer For example, 7 and 5.	12 Computed in Python.

Read the given input first. Point to the fact or value needed for the requested output.

Run the matching example, then try the fresh input at the end of the notebook. Compare both results with the known input.

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Choose the setting or tool for its job

A different choice changes the work. These are the choices we compare.

Direct call

We choose the function.

Good for testing it first.

Model request

The model proposes a call.

Inspect it before running.

Allowed tools

Expose only needed tools.

Do not grant all functions.

Define add_numbers with two typed integer inputs and a clear description.

Inspect its name and schema, then invoke it directly.

> Try different integers and a wrong input type. Check which values the schema accepts.

Practice, check, and read the full lesson

Use the notebook to try the steps. Use the article when you need the longer explanation.

1. Build

Run the small example.

Inspect each step.

2. Change

Try the fresh input.

Keep the goal clear.

3. Check

Compare with the source.

Keep mistakes visible.

Try different integers and a wrong input type. Check which values the schema accepts.

Read the complete tutorial breakdown for this section in our KGP Talkie article.

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DAY 36 · COMPLETE

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